Submission deadline and procedure: not assessed.

Introduction

This is not an assessed exercise. You will be given assistance with the tools and with coding problems. Feedback and assistance will be provided over the course of the lab session. Sample solutions will be discussed, and further feedback will be provided, as appropriate.

Resources

The following texts are available from Blackboard in electronic (pdf) form:

- Using Z, Woodcock and Davies
- Theory and practice of concurrency, Roscoe
- Concurrent and real-time system, Schneider

Each of these texts are available for your personal use and reference only.

You may find Woodcock and Davies a useful study resource to help you with task 1. For tasks 2 and 3, you may find Schneider, and Roscoe, useful resources.

Background

In the lectures, we have discussed the concept of semantics: that is, how to calculate the precise meaning of a process. We explored two types of semantics: algebraic (the use of algebraic laws to prove that two entities are related) and denotational (the use of another notation – in our case sequences of events – to define the meaning of constructs). This then allowed us to build the concept of a specification – a precise description of something a process (or system!) should, or should not, be allowed to do. The ability to construct specifications and to reason about them is fundamental to the verification process. In this lab exercise, you will re-invigorate your understanding of sets and sequences, hone your skills at the application of algebraic laws to CSP processes, and demonstrate the fine art of constructing specifications!

This lab exercise is a paper exercise – although you might find it helpful to investigate some examples using FDR and Probe, we have not yet covered sufficient material to understand how the tools could be helpful here. Therefore you should consider your answers to each question as a pen and paper exercise, although you may also find it useful to type up your solutions as a word document.

Task 1

Sets and sequences. You should answer the following questions, using standard (ZF) set theory notation.

1. Define (by extension) the set of small prime numbers, where a "small prime number" means it is less than 20.

2. Set comprehension: one of the simplest ways to create a new set is to take an existing set and to filter its members through some predicate. For some set S and predicate P, this has the general form:

   $\{x{:}S \mid P(x)\}$

Submission deadline and procedure: not assessed.

Assume the predicate Prime(x) is true when the variable x is prime. Further assume we have the set of natural numbers $\mathbb{N}$. Re-write your answer to question 1 above in this style.

3. Assume we have the set

   S == {a, b, c}

   Give the powerset of S, denoted $\mathbb{P}(S)$
   Hint: remember that for any given set X, both {} and X are subsets of X!

4. Assume we have the sets

   T == {a, b, c}
   U == {c, d, e}

   Give the union of the two sets, denoted $T \cup U$
   Give the intersection of the two sets, denoted $T \cap U$
   Give the difference of the two sets, denoted $T \setminus U$

5. The ticket office in Derby Station has a choice of two counters at which tickets may be purchased. There are two queues, one at each counter, and we can model each queue as a sequence:

   QueueA == < sally, tim, ulla >
   QueueB == < vicky, wilson, xavier >

   Just as Vicky is about to be served, counter B breaks down and the people there are asked to join the queue at counter A, meaning that counter A has queue QueueA ^ QueueB. Expand this queue!

6. Give the result of the following operation:

   $\langle a, b, c, d, e, d, c, b, a \rangle \upharpoonright^{r} \{a, d\}$

7. In Derby Station, there is a destination board displaying the trains in order of departure times. This may be modelled as a sequence of pairs, each recording a time and a destination:

   trains == < (10.15, London), (10:38, Leicester), (10:40, London),
   (11:15, Birmingham), (11:20, Nottingham), (11:40, London) >

   In order to simplify planning her journey, Sally has decided to filter the information presented to her as follows:

   trains $\upharpoonright^{r}$ { t : Time • (t, London) }

   What information is Sally considering?

   Charlie is a chatty guy, and, to stimulate conversation, asks Sally

   # (trains $\upharpoonright^{r}$ { t : Time • (t, London) })

   Assuming Sally wishes to answer Charlie politely and correctly, what answer does Sally give, and why?

Submission deadline and procedure: not assessed.

8. Sally has planned her journey, and chosen the following train:

   head(trains $|^{r}$ { t : Time ● (t, London))

   Which train has Sally selected?

9. Charlie wanted to join Sally, but was delayed buying a ticket because he was chatting too much and has missed the first train.

   Describe all the possible trains left available to Charlie. How may this description be restricted only to trains destined for London?

10. Finally, which of each of the following are equivalent, or not equivalent?
    a. $\langle a, b, c \rangle$ and $\langle a, b, c \rangle$
    b. $\langle a, b, c, c \rangle$ and $\langle a, b, c \rangle$
    c. $\{c, a, b\}$ and $\{a, b, c\}$
    d. $\{a, b, c\}$ and $\{a, b, c, c\}$
    e. $\mathbb{P}(\{a, b, c\})$ and $\{a, b, c, \{\}\}$

Task 2

1. State which of the following are true, and which are false:
   a. $a \rightarrow STOP \;\square\; b \rightarrow STOP = b \rightarrow STOP \;\square\; a \rightarrow STOP$
   b. $a \rightarrow STOP \;\sqcap\; b \rightarrow STOP = a \sqcap b \rightarrow STOP$
   c. $(a \rightarrow STOP \;\square\; b \rightarrow STOP) \;\square\; c \rightarrow STOP$
      $= a \rightarrow STOP \;\square\; (b \rightarrow STOP \;\square\; c \rightarrow STOP)$
   d. $a \rightarrow P \;\square\; a \rightarrow P = a \rightarrow P$
   e. $(a \rightarrow STOP \;\square\; b \rightarrow STOP) \sqcap c \rightarrow STOP$
      $= a \rightarrow STOP \;\square\; (b \rightarrow STOP \sqcap c \rightarrow STOP)$
   f. $a \rightarrow (P \sqcap Q) = (a \rightarrow Q) \sqcap (a \rightarrow P)$
   g. if $P = a \rightarrow P$ and $Q = RUN_{\{a\}}$ then P is not equal to Q
   h. $STOP \;\square\; P = STOP$

2. Give the traces of the following processes:
   a. $a \rightarrow STOP \;\square\; b \rightarrow c \rightarrow STOP$
   b. $a \rightarrow STOP \;\square\; b \rightarrow (c \rightarrow STOP \;\square\; d \rightarrow STOP)$
   c. $a \rightarrow STOP \;\square\; a \rightarrow b \rightarrow STOP$
   d. $a \rightarrow b \rightarrow RUN_{\{a,b\}}$
   e. $a \rightarrow b \rightarrow STOP \sqcap RUN_{\{a,c\}}$

Task 3

1. In the lecture we constructed a specification of hygiene handling raw meat. Identify which of the following are true, assuming the definitions given in the lecture.

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a. $\mathrm{RUN}_{\{\mathrm{raw,\ washed,\ cooked}\}} \sqsubseteq_T \mathrm{Spec}_1$
b. $\mathrm{Spec}_1 \sqsubseteq_T \mathrm{RUN}_\Sigma$
c. $\mathrm{STOP} \sqsubseteq_T \mathrm{Spec}_2$
d. $\mathrm{Spec}_2 \sqsubseteq_T \mathrm{STOP}$
e. $\mathrm{Spec}_1 \sqsubseteq_T \mathrm{Spec}_2$
f. $\mathrm{Spec}_2 \sqsubseteq_T \mathrm{Spec}_1$

2. Assume a list has the alphabet {open, close, moving, stopped}. Construct a specification that says the lists doors should not open whilst it is moving.
3. What does the predicate $tr \leq \langle a, b, c\rangle^\frown tr$ specify?
4. Assume a buffer process has the alphabet {out, in} where the channels are used to communicate natural numbers. Write a specification that says when a given value is output after it has been input, the buffered value should not change, both as a property specification and as a CSP process.
5. Assume another buffer process has as part of it's alphabet {write, engage, release}. Write a specification that says write should always occur between release and engage, both as a property specification and a CSP process.

Submission

You are not required to submit your work for this exercise as it is not an assessed exercise. However you should make sure you have attempted all of the problems, especially the examples on sets, sequences, and logic, as well as the algebraic laws, as you will draw heavily on this expertise later in the module.